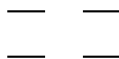


1 Orbitals And Their Symmetry

We use the following ordering of orbitals here (assuming degeneracy of LUMOs and HOMOs):

$$\begin{array}{cc} b_{1u} & \text{---} & \text{---} & a_u \\ b_{2g} & \text{---} & \text{---} & b_{3g} \end{array}$$

In the following, we will leave out the explicit labeling by orbital symmetry. Each monomer will simply be written as:



where the character of the orbital is defined by its position.

Dimer configurations will be written as their monomer occupations, by writing one monomer to the left and one to the right. The orbital ordering within the group of monomer orbitals follows the above mentioned definition.

The transformation of monomer orbitals into dimer orbitals is given by knowing the negative and positive linear combinations of monomer orbitals into dimer orbitals. Transforming this known equation system leads to:

$$\begin{aligned} a_u^l &= +b_{1g} + a_u \\ b_{1u}^l &= +a_g + b_{1u} \\ b_{2g}^l &= +b_{3u} + b_{2g} \\ b_{3g}^l &= +b_{2u} + b_{3g} \\ a_u^r &= -b_{1g} + a_u \\ b_{1u}^r &= -a_g + b_{1u} \\ b_{2g}^r &= -b_{3u} + b_{2g} \\ b_{3g}^r &= -b_{2u} + b_{3g} \end{aligned}$$

2 Monomer States and Configurations

$$S = + \begin{pmatrix} \overline{} & \overline{} \\ \uparrow\uparrow & \uparrow\uparrow \end{pmatrix} = |0220|$$

$$Q = + \begin{pmatrix} \uparrow & \uparrow \\ \uparrow & \uparrow \end{pmatrix} = |aaaa|$$

$$i^3 b_{2u} = + \begin{pmatrix} \uparrow & \overline{} \\ \uparrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} \overline{} & \uparrow \\ \uparrow & \uparrow\uparrow \end{pmatrix} = |a2a0| - |0a2a|$$

$$e^3 b_{2u} = + \begin{pmatrix} \uparrow & \overline{} \\ \uparrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} \overline{} & \uparrow \\ \uparrow & \uparrow\uparrow \end{pmatrix} = |a2a0| + |0a2a|$$

$$e^3 b_{3u} = + \begin{pmatrix} \uparrow & \overline{} \\ \uparrow & \uparrow\uparrow \end{pmatrix} - \begin{pmatrix} \overline{} & \uparrow \\ \uparrow\uparrow & \uparrow \end{pmatrix} = |aa20| - |02aa|$$

$$i^3 b_{3u} = + \begin{pmatrix} \uparrow & \overline{} \\ \uparrow & \uparrow\uparrow \end{pmatrix} + \begin{pmatrix} \overline{} & \uparrow \\ \uparrow\uparrow & \uparrow \end{pmatrix} = |aa20| + |02aa|$$

3 Linear Combinations

$$S * Q + Q * S$$

$$+ \begin{pmatrix} \overline{\quad} & \overline{\quad} & \uparrow & \uparrow \\ \uparrow\uparrow & \uparrow\uparrow & \uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & \uparrow & \overline{\quad} & \overline{\quad} \\ \uparrow & \uparrow & \uparrow\uparrow & \uparrow\uparrow \end{pmatrix}$$

written in CI vectors:

tbi

or, expressed in terms of symmetry:

$$+2 \cdot \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right| + 2 \cdot \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| + 2 \cdot \left| \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} \right| + 2 \cdot \left| \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} \right| + 2 \cdot \left| \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} \right| + 2 \cdot \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \right|$$

$$+ 2 \cdot \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \right| + 2 \cdot \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right|$$

$$S * Q - Q * S$$

$$+ \begin{pmatrix} \overline{\quad} & \overline{\quad} & \uparrow & \uparrow \\ \uparrow\uparrow & \uparrow\uparrow & \uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & \uparrow & \overline{\quad} & \overline{\quad} \\ \uparrow & \uparrow & \uparrow\uparrow & \uparrow\uparrow \end{pmatrix}$$

written in CI vectors:

tbi

or, expressed in terms of symmetry:

$$-2 \cdot \left| \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} \right| - 2 \cdot \left| \underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}} \right| - 2 \cdot \left| \underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}} \right| - 2 \cdot \left| \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} \right| - 2 \cdot \left| \underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}} \right| - 2 \cdot \left| \underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}} \right|$$

$$- 2 \cdot \left| \underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}} \right| - 2 \cdot \left| \underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}} \right|$$

$$i^3 b_{2u} * i^3 b_{2u}$$

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow\uparrow & \uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow\uparrow & \uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & \uparrow & - \\ -\uparrow & \uparrow\uparrow & \uparrow\uparrow & -\uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ -\uparrow & \uparrow\uparrow & \uparrow & \uparrow\uparrow \end{pmatrix}$$

written in CI vectors:

tbi

or, expressed in terms of symmetry:

$$\begin{aligned} & + \left| \underbrace{a_g a_g b_{2u} b_{2u}}_{a_g} \right| - \left| \underbrace{a_g a_g b_{3g} b_{3g}}_{a_g} \right| - \left| \underbrace{b_{2u} b_{2u} b_{1u} b_{1u}}_{a_g} \right| + \left| \underbrace{b_{1u} b_{1u} b_{3g} b_{3g}}_{a_g} \right| - 2 \cdot \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right| - 2 \cdot \left| \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} \right| \\ & + 2 \cdot \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| + 2 \cdot \left| \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} \right| + 2 \cdot \left| \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} \right| + 2 \cdot \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \right| - 2 \cdot \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \right| - 2 \cdot \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right| \\ & + \left| \underbrace{b_{3u} b_{3u} b_{1g} b_{1g}}_{a_g} \right| - \left| \underbrace{b_{3u} b_{3u} a_u a_u}_{a_g} \right| - \left| \underbrace{b_{1g} b_{1g} b_{2g} b_{2g}}_{a_g} \right| + \left| \underbrace{b_{2g} b_{2g} a_u a_u}_{a_g} \right| \end{aligned}$$

$$i^3 b_{2u} * e^3 b_{2u} + e^3 b_{2u} * i^3 b_{2u}$$

$$\begin{aligned} & + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow\uparrow & \uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow\uparrow & \uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow\uparrow & \uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ -\uparrow & \uparrow\uparrow & \uparrow\uparrow & -\uparrow \end{pmatrix} \\ & - \begin{pmatrix} - & \uparrow & \uparrow & - \\ -\uparrow & \uparrow\uparrow & \uparrow\uparrow & -\uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow\uparrow & \uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ -\uparrow & \uparrow\uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ -\uparrow & \uparrow\uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} \end{aligned}$$

written in CI vectors:

tbi

or, expressed in terms of symmetry:

$$\begin{aligned} & + 2 \cdot \left| \underbrace{a_g a_g b_{2u} b_{2u}}_{a_g} \right| - 2 \cdot \left| \underbrace{a_g a_g b_{3g} b_{3g}}_{a_g} \right| - 2 \cdot \left| \underbrace{b_{2u} b_{2u} b_{1u} b_{1u}}_{a_g} \right| + 2 \cdot \left| \underbrace{b_{1u} b_{1u} b_{3g} b_{3g}}_{a_g} \right| - 2 \cdot \left| \underbrace{b_{3u} b_{3u} b_{1g} b_{1g}}_{a_g} \right| + 2 \cdot \left| \underbrace{b_{3u} b_{3u} a_u a_u}_{a_g} \right| \\ & + 2 \cdot \left| \underbrace{b_{1g} b_{1g} b_{2g} b_{2g}}_{a_g} \right| - 2 \cdot \left| \underbrace{b_{2g} b_{2g} a_u a_u}_{a_g} \right| \end{aligned}$$

$$i^3 b_{2u} * e^3 b_{2u} - e^3 b_{2u} * i^3 b_{2u}$$

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow\uparrow & \uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow\uparrow & \uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow\uparrow & \uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \uparrow\uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} \\ - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \uparrow\uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow\uparrow & \uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \uparrow\uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \uparrow\uparrow & \uparrow & \uparrow\uparrow \end{pmatrix}$$

written in CI vectors:

tbi

or, expressed in terms of symmetry:

$$-4 \cdot \left| \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} \right| - 4 \cdot \left| \underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}} \right| + 4 \cdot \left| \underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}} \right| + 4 \cdot \left| \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} \right| + 4 \cdot \left| \underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}} \right| + 4 \cdot \left| \underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}} \right| \\ - 4 \cdot \left| \underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}} \right| - 4 \cdot \left| \underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}} \right|$$

$$i^3 b_{2u} * e^3 b_{3u} + e^3 b_{3u} * i^3 b_{2u}$$

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow\uparrow & \uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \uparrow\uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow\uparrow & \uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow\uparrow & \uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} \\ - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \uparrow\uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \uparrow\uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \uparrow\uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow\uparrow & \uparrow & \uparrow & \uparrow\uparrow \end{pmatrix}$$

written in CI vectors:

tbi

or, expressed in terms of symmetry:

$$+ 2 \cdot \left| \underbrace{a_g a_g b_{3u} b_{2u}}_{b_{1g}} \right| - 2 \cdot \left| \underbrace{a_g a_g b_{2g} b_{3g}}_{b_{1g}} \right| - 2 \cdot \left| \underbrace{b_{3u} b_{2u} b_{1u} b_{1u}}_{b_{1g}} \right| + 2 \cdot \left| \underbrace{b_{1u} b_{1u} b_{2g} b_{3g}}_{b_{1g}} \right| - 2 \cdot \left| \underbrace{a_g b_{2u} b_{2u} b_{1g}}_{b_{1g}} \right| + 2 \cdot \left| \underbrace{a_g b_{1g} b_{3g} b_{3g}}_{b_{1g}} \right| \\ + 2 \cdot \left| \underbrace{b_{2u} b_{2u} b_{1u} a_u}_{b_{1g}} \right| - 2 \cdot \left| \underbrace{b_{1u} b_{3g} b_{3g} a_u}_{b_{1g}} \right| - 2 \cdot \left| \underbrace{a_g b_{3u} b_{3u} b_{1g}}_{b_{1g}} \right| + 2 \cdot \left| \underbrace{b_{3u} b_{3u} b_{1u} a_u}_{b_{1g}} \right| + 2 \cdot \left| \underbrace{a_g b_{1g} b_{2g} b_{2g}}_{b_{1g}} \right| - 2 \cdot \left| \underbrace{b_{1u} b_{2g} b_{2g} a_u}_{b_{1g}} \right| \\ + 2 \cdot \left| \underbrace{b_{3u} b_{2u} b_{1g} b_{1g}}_{b_{1g}} \right| - 2 \cdot \left| \underbrace{b_{3u} b_{2u} a_u a_u}_{b_{1g}} \right| - 2 \cdot \left| \underbrace{b_{1g} b_{1g} b_{2g} b_{3g}}_{b_{1g}} \right| + 2 \cdot \left| \underbrace{b_{2g} b_{3g} a_u a_u}_{b_{1g}} \right|$$

$$i^3 b_{2u} * e^3 b_{3u} - e^3 b_{3u} * i^3 b_{2u}$$

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} \\ - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix}$$

written in CI vectors:

tbi

or, expressed in terms of symmetry:

$$-2 \cdot \left| \underbrace{a_g a_g b_{2u} b_{2g}}_{a_u} \right| + 2 \cdot \left| \underbrace{a_g a_g b_{3u} b_{3g}}_{a_u} \right| + 2 \cdot \left| \underbrace{b_{2u} b_{1u} b_{1u} b_{2g}}_{a_u} \right| - 2 \cdot \left| \underbrace{b_{3u} b_{1u} b_{1u} b_{3g}}_{a_u} \right| + 2 \cdot \left| \underbrace{a_g b_{2u} b_{2u} a_u}_{a_u} \right| - 2 \cdot \left| \underbrace{a_g b_{3g} b_{3g} a_u}_{a_u} \right| \\ -2 \cdot \left| \underbrace{b_{2u} b_{2u} b_{1g} b_{1u}}_{a_u} \right| + 2 \cdot \left| \underbrace{b_{1g} b_{1u} b_{3g} b_{3g}}_{a_u} \right| + 2 \cdot \left| \underbrace{b_{3u} b_{3u} b_{1g} b_{1u}}_{a_u} \right| - 2 \cdot \left| \underbrace{a_g b_{3u} b_{3u} a_u}_{a_u} \right| - 2 \cdot \left| \underbrace{b_{1g} b_{1u} b_{2g} b_{2g}}_{a_u} \right| + 2 \cdot \left| \underbrace{a_g b_{2g} b_{2g} a_u}_{a_u} \right| \\ -2 \cdot \left| \underbrace{b_{3u} b_{1g} b_{1g} b_{3g}}_{a_u} \right| + 2 \cdot \left| \underbrace{b_{3u} b_{3g} a_u a_u}_{a_u} \right| + 2 \cdot \left| \underbrace{b_{2u} b_{1g} b_{1g} b_{2g}}_{a_u} \right| - 2 \cdot \left| \underbrace{b_{2u} b_{2g} a_u a_u}_{a_u} \right|$$

$$i^3 b_{2u} * i^3 b_{3u} + i^3 b_{3u} * i^3 b_{2u}$$

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} \\ - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix}$$

written in CI vectors:

tbi

or, expressed in terms of symmetry:

$$+2 \cdot \left| \underbrace{a_g a_g b_{3u} b_{2u}}_{b_{1g}} \right| - 2 \cdot \left| \underbrace{a_g a_g b_{2g} b_{3g}}_{b_{1g}} \right| - 2 \cdot \left| \underbrace{b_{3u} b_{2u} b_{1u} b_{1u}}_{b_{1g}} \right| + 2 \cdot \left| \underbrace{b_{1u} b_{1u} b_{2g} b_{3g}}_{b_{1g}} \right| + 2 \cdot \left| \underbrace{a_g b_{2u} b_{2u} b_{1g}}_{b_{1g}} \right| - 2 \cdot \left| \underbrace{a_g b_{1g} b_{3g} b_{3g}}_{b_{1g}} \right| \\ -2 \cdot \left| \underbrace{b_{2u} b_{2u} b_{1u} a_u}_{b_{1g}} \right| + 2 \cdot \left| \underbrace{b_{1u} b_{3g} b_{3g} a_u}_{b_{1g}} \right| - 2 \cdot \left| \underbrace{a_g b_{3u} b_{3u} b_{1g}}_{b_{1g}} \right| + 2 \cdot \left| \underbrace{b_{3u} b_{3u} b_{1u} a_u}_{b_{1g}} \right| + 2 \cdot \left| \underbrace{a_g b_{1g} b_{2g} b_{2g}}_{b_{1g}} \right| - 2 \cdot \left| \underbrace{b_{1u} b_{2g} b_{2g} a_u}_{b_{1g}} \right| \\ -2 \cdot \left| \underbrace{b_{3u} b_{2u} b_{1g} b_{1g}}_{b_{1g}} \right| + 2 \cdot \left| \underbrace{b_{3u} b_{2u} a_u a_u}_{b_{1g}} \right| + 2 \cdot \left| \underbrace{b_{1g} b_{1g} b_{2g} b_{3g}}_{b_{1g}} \right| - 2 \cdot \left| \underbrace{b_{2g} b_{3g} a_u a_u}_{b_{1g}} \right|$$

$$i^3 b_{2u} * i^3 b_{3u} - i^3 b_{3u} * i^3 b_{2u}$$

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} \\ - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix}$$

written in CI vectors:

tbi

or, expressed in terms of symmetry:

$$-2 \cdot \left| \underbrace{a_g a_g b_{2u} b_{2g}}_{a_u} \right| + 2 \cdot \left| \underbrace{a_g a_g b_{3u} b_{3g}}_{a_u} \right| + 2 \cdot \left| \underbrace{b_{2u} b_{1u} b_{1u} b_{2g}}_{a_u} \right| - 2 \cdot \left| \underbrace{b_{3u} b_{1u} b_{1u} b_{3g}}_{a_u} \right| - 2 \cdot \left| \underbrace{a_g b_{2u} b_{2u} a_u}_{a_u} \right| + 2 \cdot \left| \underbrace{a_g b_{3g} b_{3g} a_u}_{a_u} \right| \\ + 2 \cdot \left| \underbrace{b_{2u} b_{2u} b_{1g} b_{1u}}_{a_u} \right| - 2 \cdot \left| \underbrace{b_{1g} b_{1u} b_{3g} b_{3g}}_{a_u} \right| + 2 \cdot \left| \underbrace{b_{3u} b_{3u} b_{1g} b_{1u}}_{a_u} \right| - 2 \cdot \left| \underbrace{a_g b_{3u} b_{3u} a_u}_{a_u} \right| - 2 \cdot \left| \underbrace{b_{1g} b_{1u} b_{2g} b_{2g}}_{a_u} \right| + 2 \cdot \left| \underbrace{a_g b_{2g} b_{2g} a_u}_{a_u} \right| \\ + 2 \cdot \left| \underbrace{b_{3u} b_{1g} b_{1g} b_{3g}}_{a_u} \right| - 2 \cdot \left| \underbrace{b_{3u} b_{3g} a_u a_u}_{a_u} \right| - 2 \cdot \left| \underbrace{b_{2u} b_{1g} b_{1g} b_{2g}}_{a_u} \right| + 2 \cdot \left| \underbrace{b_{2u} b_{2g} a_u a_u}_{a_u} \right|$$

$$e^3 b_{2u} * e^3 b_{2u}$$

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix}$$

written in CI vectors:

tbi

or, expressed in terms of symmetry:

$$+ \left| \underbrace{a_g a_g b_{2u} b_{2u}}_{a_g} \right| - \left| \underbrace{a_g a_g b_{3g} b_{3g}}_{a_g} \right| - \left| \underbrace{b_{2u} b_{2u} b_{1u} b_{1u}}_{a_g} \right| + \left| \underbrace{b_{1u} b_{1u} b_{3g} b_{3g}}_{a_g} \right| + 2 \cdot \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right| + 2 \cdot \left| \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} \right| \\ - 2 \cdot \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| - 2 \cdot \left| \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} \right| - 2 \cdot \left| \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} \right| - 2 \cdot \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \right| + 2 \cdot \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \right| + 2 \cdot \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right| \\ + \left| \underbrace{b_{3u} b_{3u} b_{1g} b_{1g}}_{a_g} \right| - \left| \underbrace{b_{3u} b_{3u} a_u a_u}_{a_g} \right| - \left| \underbrace{b_{1g} b_{1g} b_{2g} b_{2g}}_{a_g} \right| + \left| \underbrace{b_{2g} b_{2g} a_u a_u}_{a_g} \right|$$

$$e^3 b_{2u} * e^3 b_{3u} + e^3 b_{3u} * e^3 b_{2u}$$

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} \\ + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix}$$

written in CI vectors:

tbi

or, expressed in terms of symmetry:

$$+2 \cdot \left| \underbrace{a_g a_g b_{3u} b_{2u}}_{b_{1g}} \right| - 2 \cdot \left| \underbrace{a_g a_g b_{2g} b_{3g}}_{b_{1g}} \right| - 2 \cdot \left| \underbrace{b_{3u} b_{2u} b_{1u} b_{1u}}_{b_{1g}} \right| + 2 \cdot \left| \underbrace{b_{1u} b_{1u} b_{2g} b_{3g}}_{b_{1g}} \right| - 2 \cdot \left| \underbrace{a_g b_{2u} b_{2u} b_{1g}}_{b_{1g}} \right| + 2 \cdot \left| \underbrace{a_g b_{1g} b_{3g} b_{3g}}_{b_{1g}} \right| \\ +2 \cdot \left| \underbrace{b_{2u} b_{2u} b_{1u} a_u}_{b_{1g}} \right| - 2 \cdot \left| \underbrace{b_{1u} b_{3g} b_{3g} a_u}_{b_{1g}} \right| + 2 \cdot \left| \underbrace{a_g b_{3u} b_{3u} b_{1g}}_{b_{1g}} \right| - 2 \cdot \left| \underbrace{b_{3u} b_{3u} b_{1u} a_u}_{b_{1g}} \right| - 2 \cdot \left| \underbrace{a_g b_{1g} b_{2g} b_{2g}}_{b_{1g}} \right| + 2 \cdot \left| \underbrace{b_{1u} b_{2g} b_{2g} a_u}_{b_{1g}} \right| \\ -2 \cdot \left| \underbrace{b_{3u} b_{2u} b_{1g} b_{1g}}_{b_{1g}} \right| + 2 \cdot \left| \underbrace{b_{3u} b_{2u} a_u a_u}_{b_{1g}} \right| + 2 \cdot \left| \underbrace{b_{1g} b_{1g} b_{2g} b_{3g}}_{b_{1g}} \right| - 2 \cdot \left| \underbrace{b_{2g} b_{3g} a_u a_u}_{b_{1g}} \right|$$

$$e^3 b_{2u} * e^3 b_{3u} - e^3 b_{3u} * e^3 b_{2u}$$

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} \\ + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix}$$

written in CI vectors:

tbi

or, expressed in terms of symmetry:

$$-2 \cdot \left| \underbrace{a_g a_g b_{2u} b_{2g}}_{a_u} \right| + 2 \cdot \left| \underbrace{a_g a_g b_{3u} b_{3g}}_{a_u} \right| + 2 \cdot \left| \underbrace{b_{2u} b_{1u} b_{1u} b_{2g}}_{a_u} \right| - 2 \cdot \left| \underbrace{b_{3u} b_{1u} b_{1u} b_{3g}}_{a_u} \right| + 2 \cdot \left| \underbrace{a_g b_{2u} b_{2u} a_u}_{a_u} \right| - 2 \cdot \left| \underbrace{a_g b_{3g} b_{3g} a_u}_{a_u} \right| \\ -2 \cdot \left| \underbrace{b_{2u} b_{2u} b_{1g} b_{1u}}_{a_u} \right| + 2 \cdot \left| \underbrace{b_{1g} b_{1u} b_{3g} b_{3g}}_{a_u} \right| - 2 \cdot \left| \underbrace{b_{3u} b_{3u} b_{1g} b_{1u}}_{a_u} \right| + 2 \cdot \left| \underbrace{a_g b_{3u} b_{3u} a_u}_{a_u} \right| + 2 \cdot \left| \underbrace{b_{1g} b_{1u} b_{2g} b_{2g}}_{a_u} \right| - 2 \cdot \left| \underbrace{a_g b_{2g} b_{2g} a_u}_{a_u} \right| \\ +2 \cdot \left| \underbrace{b_{3u} b_{1g} b_{1g} b_{3g}}_{a_u} \right| - 2 \cdot \left| \underbrace{b_{3u} b_{3g} a_u a_u}_{a_u} \right| - 2 \cdot \left| \underbrace{b_{2u} b_{1g} b_{1g} b_{2g}}_{a_u} \right| + 2 \cdot \left| \underbrace{b_{2u} b_{2g} a_u a_u}_{a_u} \right|$$

$$e^3 b_{2u} * i^3 b_{3u} + i^3 b_{3u} * e^3 b_{2u}$$

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow\uparrow & \uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \uparrow\uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow\uparrow & \uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow\uparrow & \uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} \\ + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \uparrow\uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \uparrow\uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \uparrow\uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow\uparrow & \uparrow & \uparrow & \uparrow\uparrow \end{pmatrix}$$

written in CI vectors:

tbi

or, expressed in terms of symmetry:

$$+2 \cdot \left| \underbrace{a_g a_g b_{3u} b_{2u}}_{b_{1g}} \right| - 2 \cdot \left| \underbrace{a_g a_g b_{2g} b_{3g}}_{b_{1g}} \right| - 2 \cdot \left| \underbrace{b_{3u} b_{2u} b_{1u} b_{1u}}_{b_{1g}} \right| + 2 \cdot \left| \underbrace{b_{1u} b_{1u} b_{2g} b_{3g}}_{b_{1g}} \right| + 2 \cdot \left| \underbrace{a_g b_{2u} b_{2u} b_{1g}}_{b_{1g}} \right| - 2 \cdot \left| \underbrace{a_g b_{1g} b_{3g} b_{3g}}_{b_{1g}} \right| \\ - 2 \cdot \left| \underbrace{b_{2u} b_{2u} b_{1u} a_u}_{b_{1g}} \right| + 2 \cdot \left| \underbrace{b_{1u} b_{3g} b_{3g} a_u}_{b_{1g}} \right| + 2 \cdot \left| \underbrace{a_g b_{3u} b_{3u} b_{1g}}_{b_{1g}} \right| - 2 \cdot \left| \underbrace{b_{3u} b_{3u} b_{1u} a_u}_{b_{1g}} \right| - 2 \cdot \left| \underbrace{a_g b_{1g} b_{2g} b_{2g}}_{b_{1g}} \right| + 2 \cdot \left| \underbrace{b_{1u} b_{2g} b_{2g} a_u}_{b_{1g}} \right| \\ + 2 \cdot \left| \underbrace{b_{3u} b_{2u} b_{1g} b_{1g}}_{b_{1g}} \right| - 2 \cdot \left| \underbrace{b_{3u} b_{2u} a_u a_u}_{b_{1g}} \right| - 2 \cdot \left| \underbrace{b_{1g} b_{1g} b_{2g} b_{3g}}_{b_{1g}} \right| + 2 \cdot \left| \underbrace{b_{2g} b_{3g} a_u a_u}_{b_{1g}} \right|$$

$$e^3 b_{2u} * i^3 b_{3u} - i^3 b_{3u} * e^3 b_{2u}$$

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow\uparrow & \uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \uparrow\uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow\uparrow & \uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow\uparrow & \uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} \\ + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \uparrow\uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \uparrow\uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \uparrow\uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow\uparrow & \uparrow & \uparrow & \uparrow\uparrow \end{pmatrix}$$

written in CI vectors:

tbi

or, expressed in terms of symmetry:

$$-2 \cdot \left| \underbrace{a_g a_g b_{2u} b_{2g}}_{a_u} \right| + 2 \cdot \left| \underbrace{a_g a_g b_{3u} b_{3g}}_{a_u} \right| + 2 \cdot \left| \underbrace{b_{2u} b_{1u} b_{1u} b_{2g}}_{a_u} \right| - 2 \cdot \left| \underbrace{b_{3u} b_{1u} b_{1u} b_{3g}}_{a_u} \right| - 2 \cdot \left| \underbrace{a_g b_{2u} b_{2u} a_u}_{a_u} \right| + 2 \cdot \left| \underbrace{a_g b_{3g} b_{3g} a_u}_{a_u} \right| \\ + 2 \cdot \left| \underbrace{b_{2u} b_{2u} b_{1g} b_{1u}}_{a_u} \right| - 2 \cdot \left| \underbrace{b_{1g} b_{1u} b_{3g} b_{3g}}_{a_u} \right| - 2 \cdot \left| \underbrace{b_{3u} b_{3u} b_{1g} b_{1u}}_{a_u} \right| + 2 \cdot \left| \underbrace{a_g b_{3u} b_{3u} a_u}_{a_u} \right| + 2 \cdot \left| \underbrace{b_{1g} b_{1u} b_{2g} b_{2g}}_{a_u} \right| - 2 \cdot \left| \underbrace{a_g b_{2g} b_{2g} a_u}_{a_u} \right| \\ - 2 \cdot \left| \underbrace{b_{3u} b_{1g} b_{1g} b_{3g}}_{a_u} \right| + 2 \cdot \left| \underbrace{b_{3u} b_{3g} a_u a_u}_{a_u} \right| + 2 \cdot \left| \underbrace{b_{2u} b_{1g} b_{1g} b_{2g}}_{a_u} \right| - 2 \cdot \left| \underbrace{b_{2u} b_{2g} a_u a_u}_{a_u} \right|$$

$$e^3 b_{3u} * e^3 b_{3u}$$

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \uparrow\uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \uparrow\uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow\uparrow & \uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow\uparrow & \uparrow & \uparrow\uparrow & \uparrow \end{pmatrix}$$

written in CI vectors:

tbi

or, expressed in terms of symmetry:

$$\begin{aligned} & + \left| \underbrace{a_g a_g b_{3u} b_{3u}}_{a_g} \right| - \left| \underbrace{a_g a_g b_{2g} b_{2g}}_{a_g} \right| - \left| \underbrace{b_{3u} b_{3u} b_{1u} b_{1u}}_{a_g} \right| + \left| \underbrace{b_{1u} b_{1u} b_{2g} b_{2g}}_{a_g} \right| - 2 \cdot \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right| - 2 \cdot \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| \\ & + 2 \cdot \left| \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} \right| + 2 \cdot \left| \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} \right| + 2 \cdot \left| \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} \right| + 2 \cdot \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \right| - 2 \cdot \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \right| - 2 \cdot \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right| \\ & + \left| \underbrace{b_{2u} b_{2u} b_{1g} b_{1g}}_{a_g} \right| - \left| \underbrace{b_{2u} b_{2u} a_u a_u}_{a_g} \right| - \left| \underbrace{b_{1g} b_{1g} b_{3g} b_{3g}}_{a_g} \right| + \left| \underbrace{b_{3g} b_{3g} a_u a_u}_{a_g} \right| \end{aligned}$$

$$e^3 b_{3u} * i^3 b_{3u} + i^3 b_{3u} * e^3 b_{3u}$$

$$\begin{aligned} & + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \uparrow\uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \uparrow\uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \uparrow\uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow\uparrow & \uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} \\ & - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow\uparrow & \uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \uparrow\uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow\uparrow & \uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow\uparrow & \uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} \end{aligned}$$

written in CI vectors:

tbi

or, expressed in terms of symmetry:

$$\begin{aligned} & + 2 \cdot \left| \underbrace{a_g a_g b_{3u} b_{3u}}_{a_g} \right| - 2 \cdot \left| \underbrace{a_g a_g b_{2g} b_{2g}}_{a_g} \right| - 2 \cdot \left| \underbrace{b_{3u} b_{3u} b_{1u} b_{1u}}_{a_g} \right| + 2 \cdot \left| \underbrace{b_{1u} b_{1u} b_{2g} b_{2g}}_{a_g} \right| - 2 \cdot \left| \underbrace{b_{2u} b_{2u} b_{1g} b_{1g}}_{a_g} \right| + 2 \cdot \left| \underbrace{b_{2u} b_{2u} a_u a_u}_{a_g} \right| \\ & + 2 \cdot \left| \underbrace{b_{1g} b_{1g} b_{3g} b_{3g}}_{a_g} \right| - 2 \cdot \left| \underbrace{b_{3g} b_{3g} a_u a_u}_{a_g} \right| \end{aligned}$$

$$e^3 b_{3u} * i^3 b_{3u} - i^3 b_{3u} * e^3 b_{3u}$$

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \uparrow\uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \uparrow\uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \uparrow\uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow\uparrow & \uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} \\ - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow\uparrow & \uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \uparrow\uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow\uparrow & \uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow\uparrow & \uparrow & \uparrow\uparrow & \uparrow \end{pmatrix}$$

written in CI vectors:

tbi

or, expressed in terms of symmetry:

$$-4 \cdot \left| \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} \right| - 4 \cdot \left| \underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}} \right| + 4 \cdot \left| \underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}} \right| + 4 \cdot \left| \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} \right| + 4 \cdot \left| \underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}} \right| + 4 \cdot \left| \underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}} \right| \\ - 4 \cdot \left| \underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}} \right| - 4 \cdot \left| \underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}} \right|$$

$$i^3 b_{3u} * i^3 b_{3u}$$

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \uparrow\uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \uparrow\uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow\uparrow & \uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow\uparrow & \uparrow & \uparrow\uparrow & \uparrow \end{pmatrix}$$

written in CI vectors:

tbi

or, expressed in terms of symmetry:

$$+ \left| \underbrace{a_g a_g b_{3u} b_{3u}}_{a_g} \right| - \left| \underbrace{a_g a_g b_{2g} b_{2g}}_{a_g} \right| - \left| \underbrace{b_{3u} b_{3u} b_{1u} b_{1u}}_{a_g} \right| + \left| \underbrace{b_{1u} b_{1u} b_{2g} b_{2g}}_{a_g} \right| + 2 \cdot \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right| + 2 \cdot \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| \\ - 2 \cdot \left| \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} \right| - 2 \cdot \left| \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} \right| - 2 \cdot \left| \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} \right| - 2 \cdot \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \right| + 2 \cdot \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \right| + 2 \cdot \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right| \\ + \left| \underbrace{b_{2u} b_{2u} b_{1g} b_{1g}}_{a_g} \right| - \left| \underbrace{b_{2u} b_{2u} a_u a_u}_{a_g} \right| - \left| \underbrace{b_{1g} b_{1g} b_{3g} b_{3g}}_{a_g} \right| + \left| \underbrace{b_{3g} b_{3g} a_u a_u}_{a_g} \right|$$

4 Symmetry Conclusion

Dimer State	Symmetry
$S * Q + Q * S$	a_g
$S * Q - Q * S$	b_{1u}
$i^3 b_{2u} * i^3 b_{2u}$	a_g
$i^3 b_{2u} * e^3 b_{2u} + e^3 b_{2u} * i^3 b_{2u}$	a_g
$i^3 b_{2u} * e^3 b_{2u} - e^3 b_{2u} * i^3 b_{2u}$	b_{1u}
$i^3 b_{2u} * e^3 b_{3u} + e^3 b_{3u} * i^3 b_{2u}$	b_{1g}
$i^3 b_{2u} * e^3 b_{3u} - e^3 b_{3u} * i^3 b_{2u}$	a_u
$i^3 b_{2u} * i^3 b_{3u} + i^3 b_{3u} * i^3 b_{2u}$	b_{1g}
$i^3 b_{2u} * i^3 b_{3u} - i^3 b_{3u} * i^3 b_{2u}$	a_u
$e^3 b_{2u} * e^3 b_{2u}$	a_g
$e^3 b_{2u} * e^3 b_{3u} + e^3 b_{3u} * e^3 b_{2u}$	b_{1g}
$e^3 b_{2u} * e^3 b_{3u} - e^3 b_{3u} * e^3 b_{2u}$	a_u
$e^3 b_{2u} * i^3 b_{3u} + i^3 b_{3u} * e^3 b_{2u}$	b_{1g}
$e^3 b_{2u} * i^3 b_{3u} - i^3 b_{3u} * e^3 b_{2u}$	a_u
$e^3 b_{3u} * e^3 b_{3u}$	a_g
$e^3 b_{3u} * i^3 b_{3u} + i^3 b_{3u} * e^3 b_{3u}$	a_g
$e^3 b_{3u} * i^3 b_{3u} - i^3 b_{3u} * e^3 b_{3u}$	b_{1u}
$i^3 b_{3u} * i^3 b_{3u}$	a_g

$\Rightarrow$ 7x a_g and 3x b_{1u} and 4x b_{1g} and 4x a_u